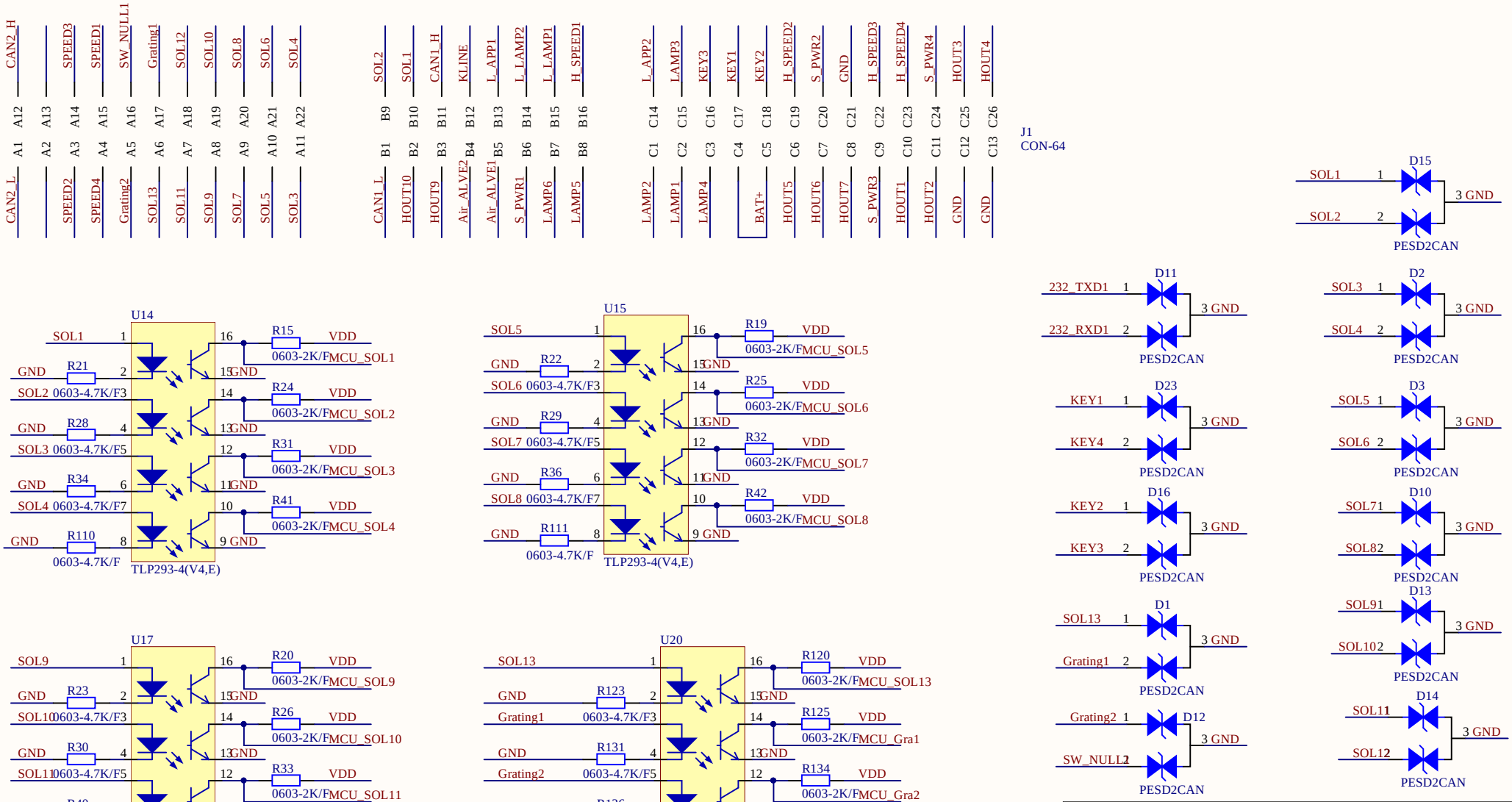
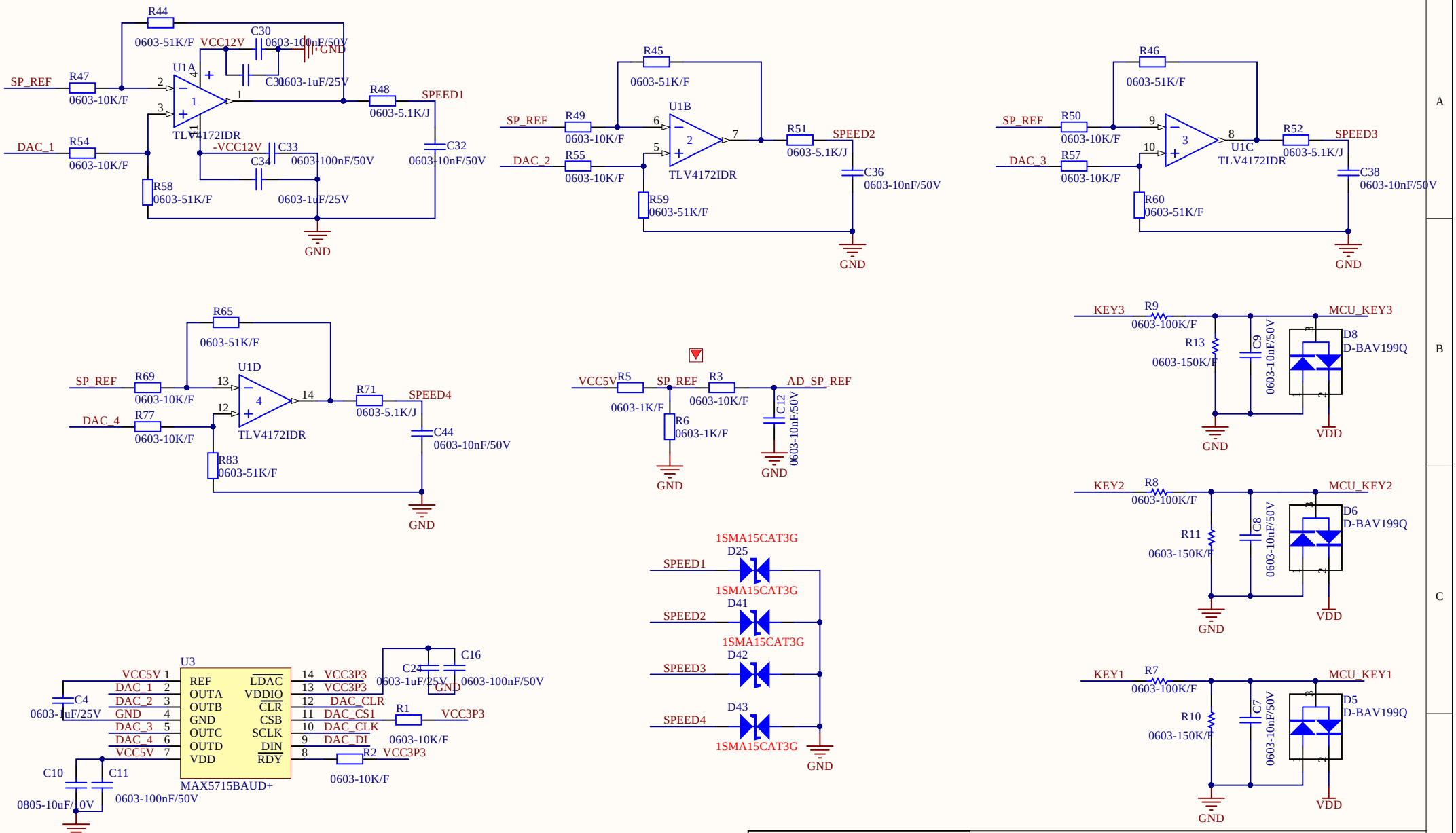
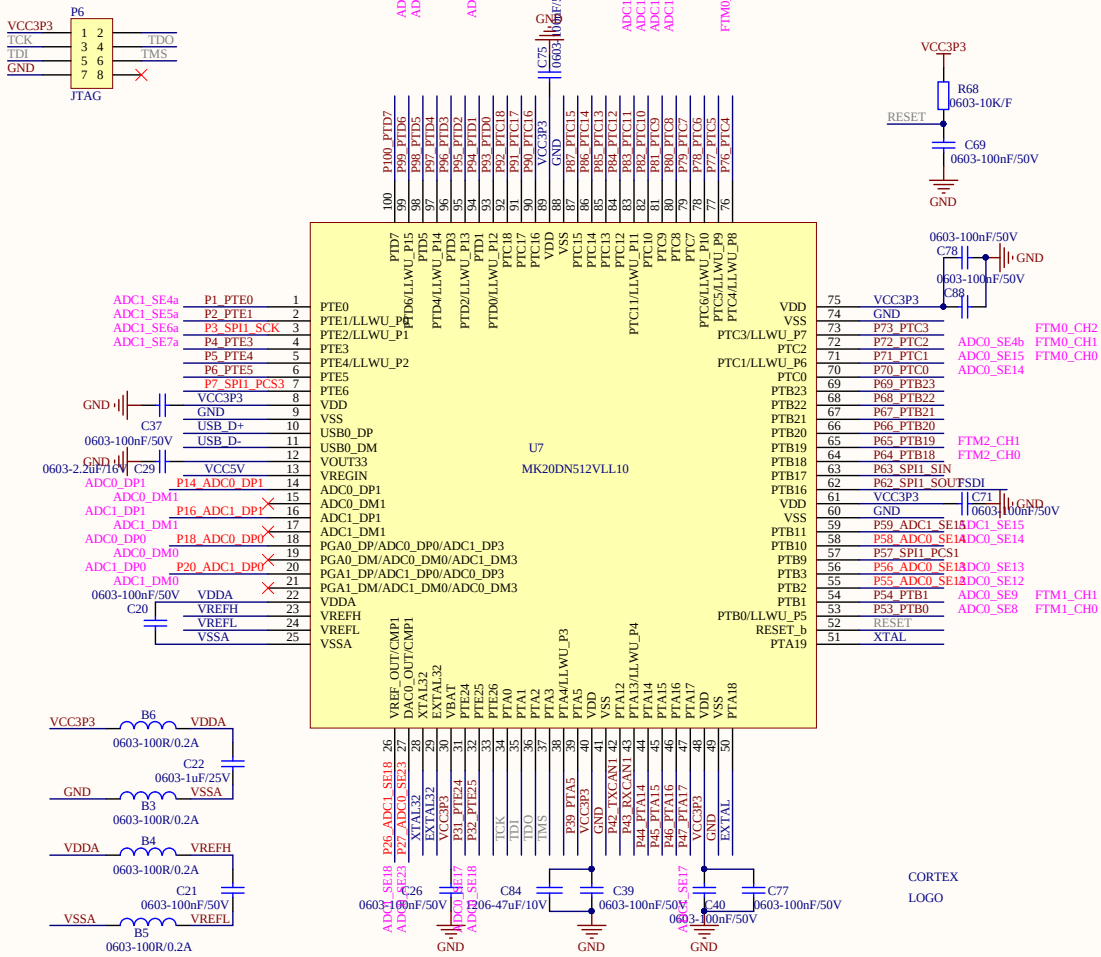
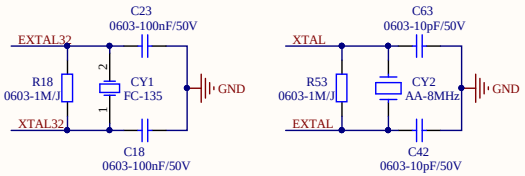
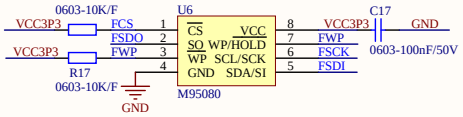

J1
CON-64
CAN2_H
CAN2_L
SPEED3
SPEED2
SPEED1
SPEED4
SW_NULL1
Grating1
Grating2
SOL12
SOL13
SOL10
SOL11
SOL8
SOL9
SOL6
SOL7
SOL4
SOL5
SOL3
SOL2
CAN1_L
SOL1
HOUT10
CAN1_H
HOUT9
KLINE
Air_ALVE2
L_APP1
Air_ALVE1
L_LAMP2
S_PWR1
L_LAMP1
LAMP6
H_SPEED1
LAMP5
L_APP2
LAMP2
LAMP3
LAMP1
KEY3
LAMP4
KEY1
BAT+
KEY2
HOUT5
H_SPEED2
HOUT6
S_PWR2
HOUT7
GND
S_PWR3
H_SPEED3
HOUT1
H_SPEED4
HOUT2
S_PWR4
GND
HOUT3
GND
HOUT4
U14
TLP293-4(V4,E)
R15 0603-2K/F VDD MCU_SOL1
R24 0603-2K/F VDD MCU_SOL2
R31 0603-2K/F VDD MCU_SOL3
R41 0603-2K/F VDD MCU_SOL4
R21 0603-4.7K/F
R28 0603-4.7K/F
R34 0603-4.7K/F
R110 0603-4.7K/F
U15
TLP293-4(V4,E)
R19 0603-2K/F VDD MCU_SOL5
R25 0603-2K/F VDD MCU_SOL6
R32 0603-2K/F VDD MCU_SOL7
R42 0603-2K/F VDD MCU_SOL8
R22 0603-4.7K/F
R29 0603-4.7K/F
R36 0603-4.7K/F
R111 0603-4.7K/F
U17
TLP293-4(V4,E)
R20 0603-2K/F VDD MCU_SOL9
R26 0603-2K/F VDD MCU_SOL10
R33 0603-2K/F VDD MCU_SOL11
R43 0603-2K/F VDD MCU_SOL12
R23 0603-4.7K/F
R30 0603-4.7K/F
R40 0603-4.7K/F
R112 0603-4.7K/F
U20
TLP293-4(V4,E)
R120 0603-2K/F VDD MCU_SOL13
R125 0603-2K/F VDD MCU_Gra1
R134 0603-2K/F VDD MCU_Gra2
R139 0603-2K/F VDD MCU_NULL1
R123 0603-4.7K/F
R131 0603-4.7K/F
R136 0603-4.7K/F
R141 0603-4.7K/F
D15 PESD2CAN
D11 PESD2CAN
D2 PESD2CAN
D23 PESD2CAN
D3 PESD2CAN
D16 PESD2CAN
D10 PESD2CAN
D1 PESD2CAN
D13 PESD2CAN
D12 PESD2CAN
D14 PESD2CAN
3 GND
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R44
0603-51K/F
VCC12V
C30
0603-100nF/50V
GND
C30
0603-1uF/25V
SP_REF
R47
0603-10K/F
U1A
TLV4172IDR
R48
0603-5.1K/J
SPEED1
DAC_1
R54
0603-10K/F
-VCC12V
C33
C34
0603-100nF/50V
C32
0603-10nF/50V
R58
0603-51K/F
0603-1uF/25V
GND
R45
0603-51K/F
SP_REF
R49
0603-10K/F
U1B
TLV4172IDR
R51
0603-5.1K/J
SPEED2
DAC_2
R55
0603-10K/F
R59
0603-51K/F
C36
0603-10nF/50V
GND
R46
0603-51K/F
SP_REF
R50
0603-10K/F
U1C
TLV4172IDR
R52
0603-5.1K/J
SPEED3
DAC_3
R57
0603-10K/F
R60
0603-51K/F
C38
0603-10nF/50V
GND
R65
0603-51K/F
SP_REF
R69
0603-10K/F
U1D
TLV4172IDR
R71
0603-5.1K/J
SPEED4
DAC_4
R77
0603-10K/F
R83
0603-51K/F
C44
0603-10nF/50V
GND
VCC5V
R5
0603-1K/F
SP_REF
R3
0603-10K/F
AD_SP_REF
R6
0603-1K/F
C12
0603-10nF/50V
GND
KEY3
R9
0603-100K/F
MCU_KEY3
R13
0603-150K/F
C9
0603-10nF/50V
D8
D-BAV199Q
GND
VDD
KEY2
R8
0603-100K/F
MCU_KEY2
R11
0603-150K/F
C8
0603-10nF/50V
D6
D-BAV199Q
GND
VDD
KEY1
R7
0603-100K/F
MCU_KEY1
R10
0603-150K/F
C7
0603-10nF/50V
D5
D-BAV199Q
GND
VDD
1SMA15CAT3G
D25
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D41
SPEED2
1SMA15CAT3G
D42
SPEED3
1SMA15CAT3G
D43
SPEED4
1SMA15CAT3G
GND
U3
VCC5V 1
DAC_1 2
DAC_2 3
DAC_3 5
DAC_4 6
VCC5V 7
REF
OUTA
OUTB
GND
OUTC
OUTD
VDD
LDAC
VDDIO
CLR
CSB
SCLK
DIN
RDY
14 VCC3P3
13 VCC3P3
12 DAC_CLR
11 DAC_CS1
10 DAC_CLK
9 DAC_DI
8 R2 VCC3P3
MAX5715BAUD+
C4
0603-1uF/25V
C10
0805-10uF/10V
C11
0603-100nF/50V
C24
0603-1uF/25V
C16
0603-100nF/50V
R1
0603-10K/F
0603-10K/F
GND
MCU
DAC_DI
DAC_CLK
DAC_CS1
DAC_CS2
DAC_CLR
对外接口
GND_ECU
SPEED1
SPEED2
SPEED3
SPEED4
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SPI

P7_SPI1_PCS3		DAC_CS1
P57_SPI1_PCS1		FCS
P63_SPI1_SIN		FSDO
P3_SPI1_SCK	DAC_CLK	FSCK
P62_SPI1_SOUT	DAC_DI	FSDI

CAN

P42_TXCAN1	TXCAN1
P43_RXCAN1	RXCAN1
P91_PTC17	TXCAN2
P90_PTC16	RXCAN2

UART

P31_PTE24	KTXD
P32_PTE25	KRXD
P1_PTE0	OUT_TX1
P2_PTE1	OUT_RX1

PWM

P53_PTB0	FTM1_CH0	SOL1IN
P72_PTC2	FTM0_CH1	SOL2IN
P73_PTC3	FTM0_CH2	SOL3IN
P54_PTB1	FTM1_CH1	SOL4IN
P76_PTC4	FTM0_CH3	SOL5IN
P64_PTB18	FTM2_CH0	SOL6IN
P65_PTB19	FTM2_CH1	SOL7IN
P71_PTC1	FTM0_CH0	SOL8IN

I/O

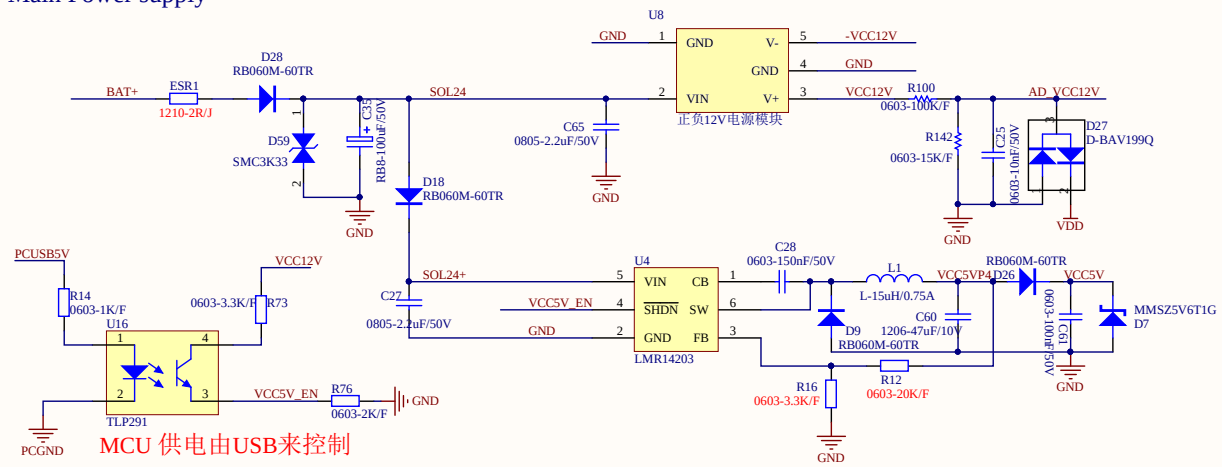
P70_PTC0	RL_DRV
P4_PTE3	MCU_Gra1
P5_PTE4	MCU_Gra2
P6_PTE5	MCU_NULL1
P44_PTA14	HALL_5VCTR
P45_PTA15	DAC_CLR
P46_PTA16	MCU_LAMP1
P47_PTA17	MCU_LAMP2
P66_PTB20	MCU_LAMP3
P67_PTB21	MCU_LAMP4
P68_PTB22	MCU_LAMP5
P69_PTB23	MCU_LAMP6
P77_PTC5	MCU_KEY3
P78_PTC6	MCU_KEY2
P79_PTC7	MCU_KEY1
P80_PTC8	SOL9IN
P81_PTC9	SOL10IN
P82_PTC10	SOL11IN
P83_PTC11	SOL12IN
P84_PTC12	MCU_SOL1
P85_PTC13	MCU_SOL2
P86_PTC14	MCU_SOL3
P87_PTC15	MCU_SOL4
P92_PTC18	MCU_SOL5
P93_PTD0	MCU_SOL6
P94_PTD1	MCU_SOL7
P95_PTD2	MCU_SOL8
P96_PTD3	MCU_SOL9
P97_PTD4	MCU_SOL10
P98_PTD5	MCU_SOL11
P99_PTD6	MCU_SOL12
P100_PTD7	MCU_SOL13
P39_PTA5	VDD2SEN

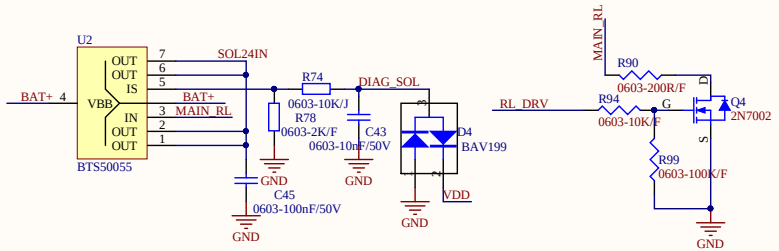
ADC

P14_ADC0_DP1	AD_MCU_SOL-current
P16_ADC1_DP1	AD_MCU_Motor_current
P18_ADC0_DP0	AD_L_APP1
P20_ADC1_DP0	AD_L_APP2
P26_ADC1_SE18	AD_VCC12V
P27_ADC0_SE23	AD_SP_REF
P55_ADC0_SE12	AD_SOLIN
P56_ADC0_SE13	AD_L_LAMP1
P58_ADC0_SE14	AD_L_LAMP2
P59_ADC1_SE15	DIAG_SOL

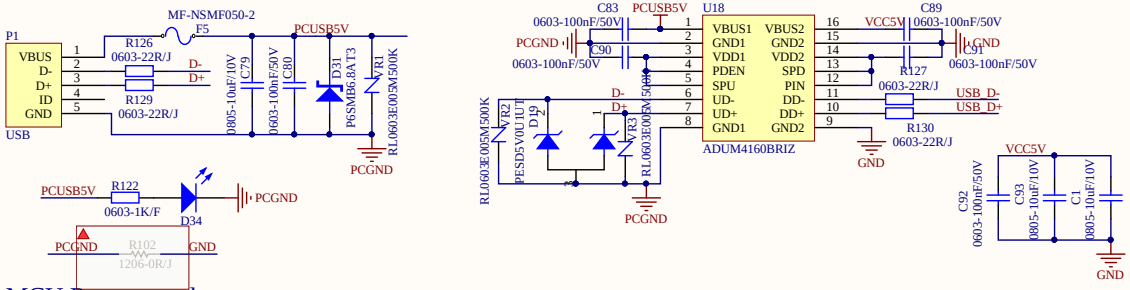
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Main Power supply

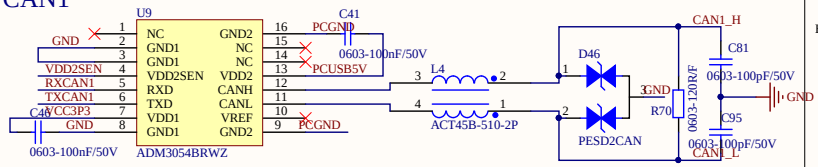




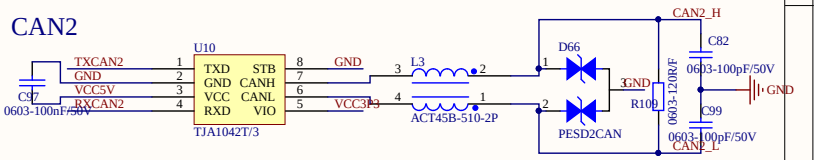
USB Power supply



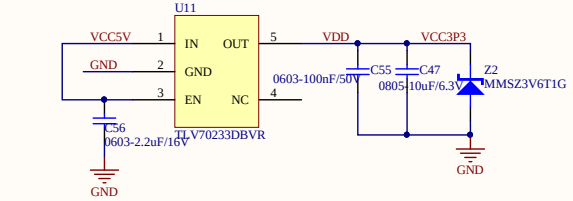
CAN1



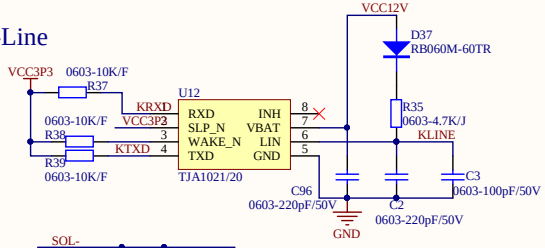
CAN2



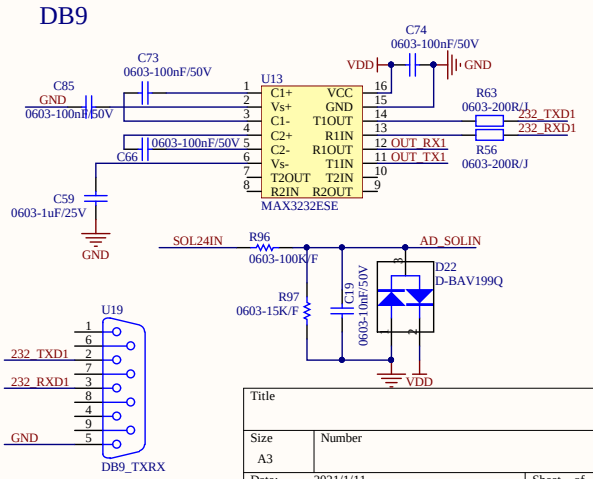
MCU Power supply

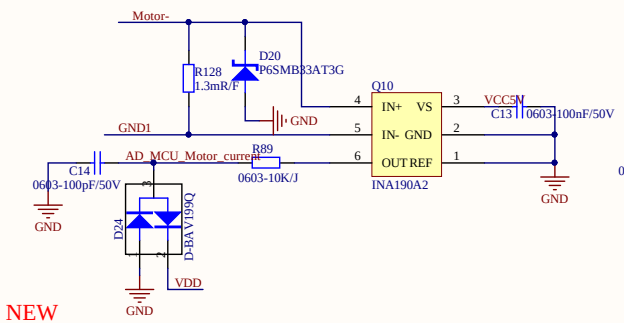


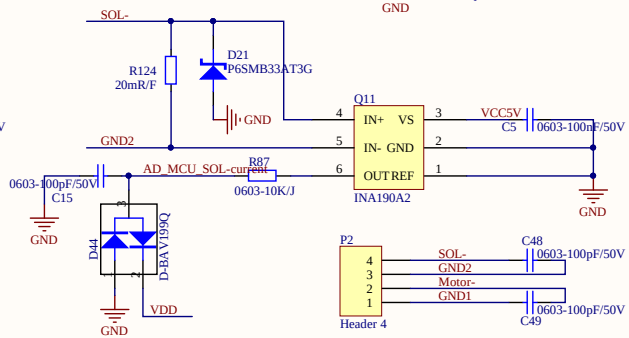
K-Line



DB9

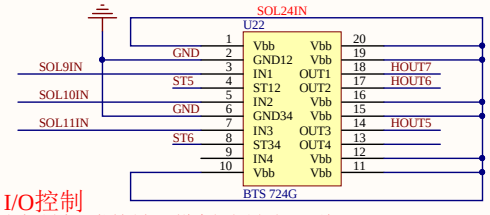






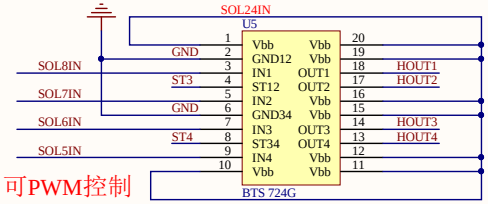
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增加供电端的I/O控制，关断MCU端12V/24V 供电

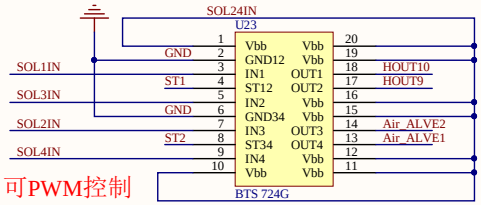


I/O控制

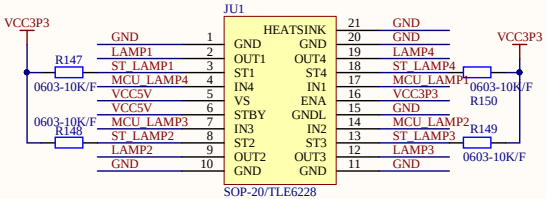
程序不改变IGN控制引脚，则线束上需要改到HOUT7端口

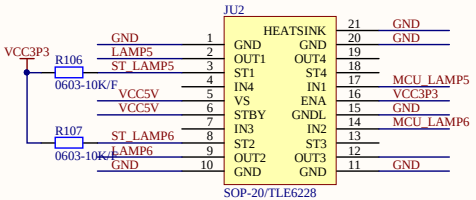


可PWM控制

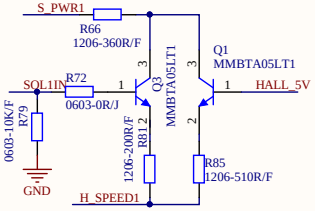


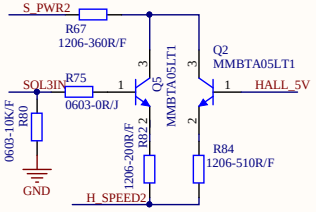
可PWM控制

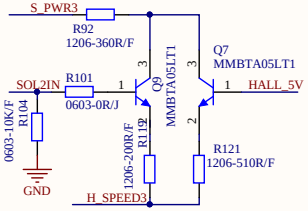


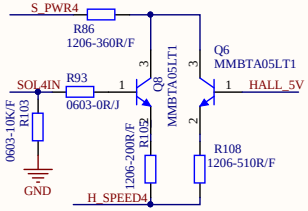


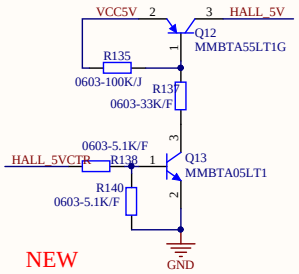
TLE6228的供电Vs和STBY由SOL24VIN改为5V







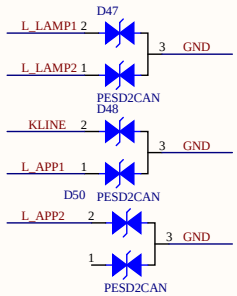


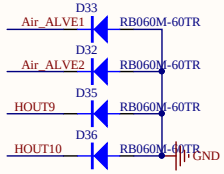


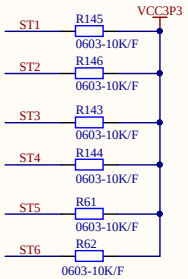
NEW

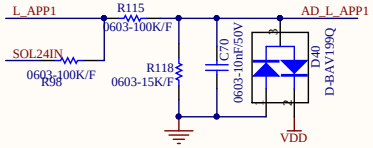
1、磁电与HALL轮速信号分别独立可控，HALL_5VCTR为低电平时，默认为HALL传感器开路状态。

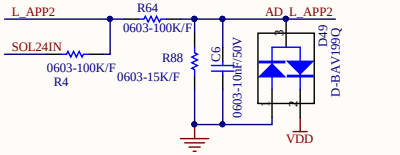
2、当测试磁传感器时，HALL信号不输出，HALL_5VCTR需拉低，ICT测试工装上可同时保留磁电与HALL接口的连接，通过下线检测软件自动切换，不需要手动修改硬线接口。

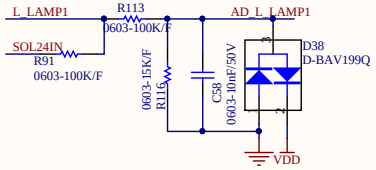


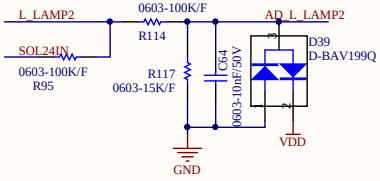












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SOL12IN